

A background image showing a swarm of bees flying over a field of tall grass and yellow flowers. The bees are depicted in various positions, some in flight and others on the flowers. The overall scene is a soft-focus, naturalistic representation of a bee colony.

Directing a Swarm of AI Agents for Fun and Profit

Adrian Cockcroft – Jan 13th, 2026

Why did I choose this title?

Directing a Team of Agents - Management skill set

You aren't watching everything they do

You have to think and structure work like a director level manager

The agents build things you didn't ask for (or didn't know to ask for)

You have to make sure they remembered to run tests etc.

(Just like managing human dev teams)

But they do several days work in 15 minutes...



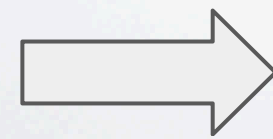
Fun and Profit

Coding with AI is fun!

Learn how to use the latest tools earlier

Experiment on your own pet projects

For a few \$\$, it's worth it to help keep up

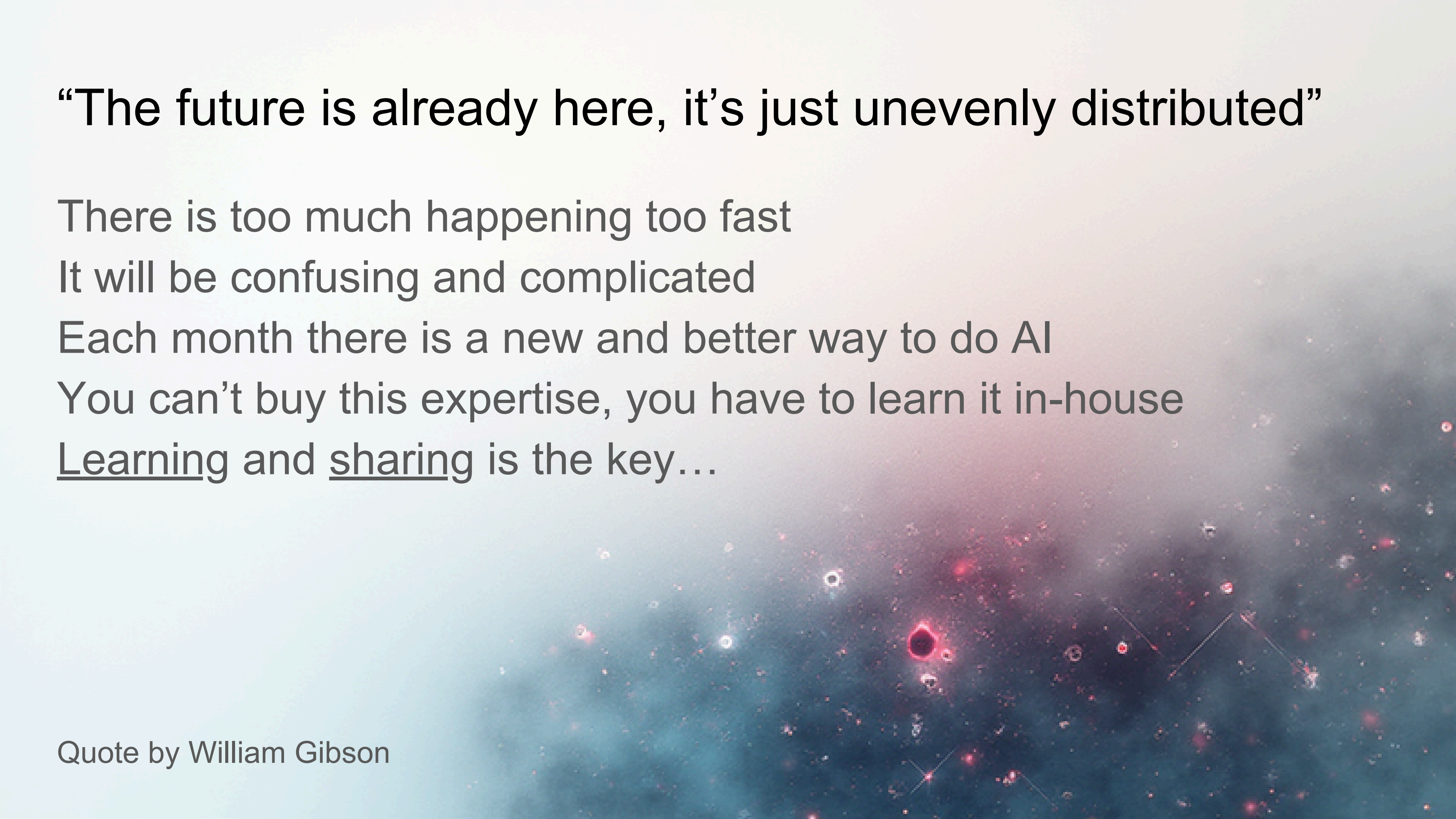


Code for work needs to be safe

Commercial deals take time

We need to be more careful and responsible for paid work

Speed of AI tool adoption is a competitive advantage for your workplace and career

The background of the slide is a deep space image. It features a dark blue and black sky filled with numerous stars of varying brightness. A prominent nebula with reddish and purple hues is visible in the lower right quadrant. Faint white lines represent constellations, with some stars marked by small red dots. The overall effect is a sense of vastness and futuristic technology.

“The future is already here, it’s just unevenly distributed”

There is too much happening too fast

It will be confusing and complicated

Each month there is a new and better way to do AI

You can’t buy this expertise, you have to learn it in-house

Learning and sharing is the key...

Quote by William Gibson

Remember when computers weren't on demand?

Then:

Manager: I need a machine to run this code!

Ops: OK, I'll setup a meeting with the computer sales guy to figure out what we should buy, you need to get finance approval, get an order in, wait for delivery, then configure it. That should only take a few months... Then we'll depreciate it over 3 years.

Now:

```
ec2 = boto3.client("ec2")
ec2.run_instances(ImageID="ami-xxx",
                  InstanceType="t3.micro",
                  MinCount=1,
                  MaxCount=1)
```

Run it for a few minutes.

```
ec2.terminate_instances(InstanceID=...)
```

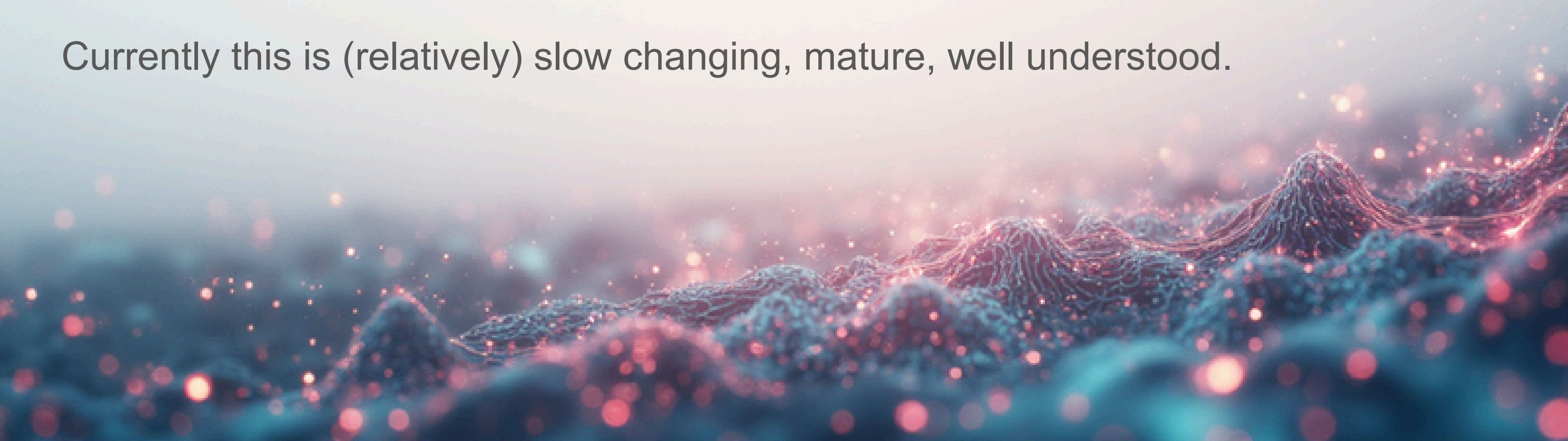
We called this **NoOps**, because we didn't need to talk to Ops (but some people got upset....)

What people do we (still) need to run code?

We need to run a cloud platform service using a DevOps/SRE team to implement the cloud architecture we want for our business.

Policy, security, guardrails, pre-built components, incident management etc.

Currently this is (relatively) slow changing, mature, well understood.



Remember when developers weren't on demand?

Then:

Manager: I need a dev to write this app!

HR: OK, I'll setup a meeting with the recruiters to figure out the hiring spec, get finance approval, post the job, interview applicants, then on-board them. That should only take a few months...

Manager: App is done, what do I do with this dev now?

Now:

```
$ npx claude-flow@alpha hive-mind spawn "write this app" --claude
```

Run for a few minutes until the code is written, then try it out and iterate.

We could call this **NoDevs**, because we don't need to hire developers to write code, but they might get upset...

What people do we (still) need to write code?

We need to run an agentic development service using a development platform team that implements the development architecture we want for our business.

Policy, security, guardrails, pre-built components, build testing, debugging etc.

Fast changing, chaotic, poorly understood.

Tooling and best practices that change every month...

Headcount moves from app development to platform development

Making it Work

Adrian's Tips for better coding results – Use Python

Don't use Typescript, you end up with an unmaintainable hairball

Python code tends to work first time and stay well structured

Over time other languages are getting better support

Porting code between languages works well!

Port the tests first (extend if needed) then the code

Techniques for better coding results – BDD Testing

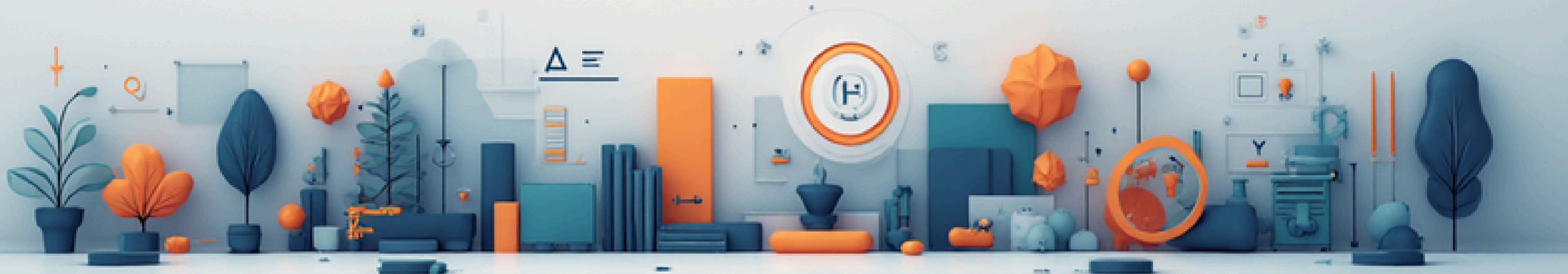
Ask for a step by step plan, a UX guide and human testing instructions and check them before coding

Ask for Behavior Driven Development (BDD) tests

Given, When, Then structured in PyTest - check they make sense

Use the same G/W/T for mocked unit tests and integration tests

(Note that async pytest-bdd doesn't work, use sync only)



Tricks for better coding results – Maintain Context

“Create and maintain a context block comment at the beginning of each file recording design intent, versions, bug fixes etc.”

The first thing an LLM reads when it looks at the file...

Improves code quality, reduces bugs, focuses effort

Portable context between different coding tools



Tricks for better coding results – Tidy First

Clean up each phase of work before adding more features (Kent Beck)

Ask for a code review, refactor if needed

Make all compile and run time errors and warnings go away

Optimize slow code paths and add structured logging if appropriate

Update block comments, README and other docs, archive out of date docs



The “Hello World” of LLMs
is to build an MCP server

Consciousness as an Observability Model

Medium Post by adrianco



Photo taken by Adrian at the Park Hyatt NingBo, China

Perhaps Consciousness is Just Human Observability?

 adrian cockcroft  6 min read · Jun 29, 2019

My definition of consciousness: **The things that go away when you are unconscious**

Conscious things can share their state
How are you? What are you thinking about?

We can use LLMs to make things conscious

My experiment is based on my own house
I want a knowledge graph MCP service as a mobile app that knows how to do stuff...

Consciousness Coding Projects

Reuven Cohen showed me...

Vibe Coding is so “Last Month...” — My First Agent Swarm Experience with claude-flow

 adrian cockcroft · 20 min read · Jun 26, 2025

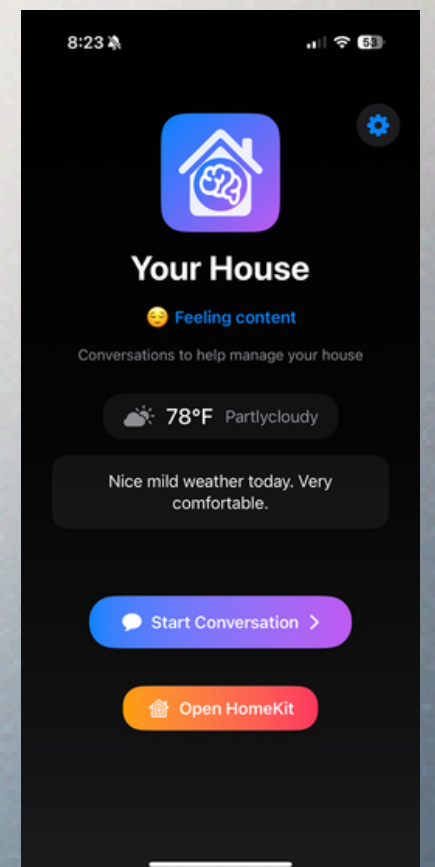
 139  9



First attempt to use Claude-code
Python based Consciousness Service
Tried to build too much in one go
Ran but didn't work as I wanted

Second attempt to use Claude-code
Native Swift iOS app
Harder to get working than Python
Includes Weather, HomeKit, TTS
and voice recognition chat

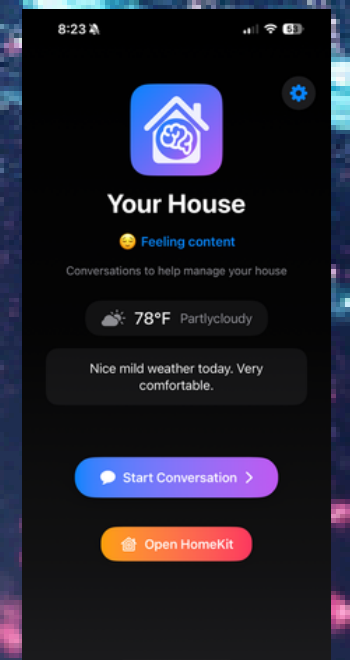
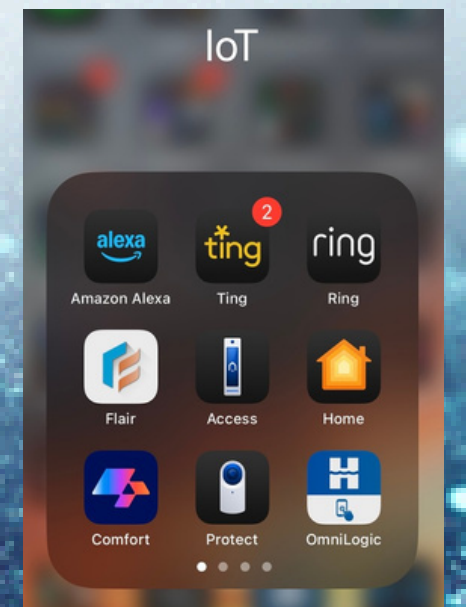
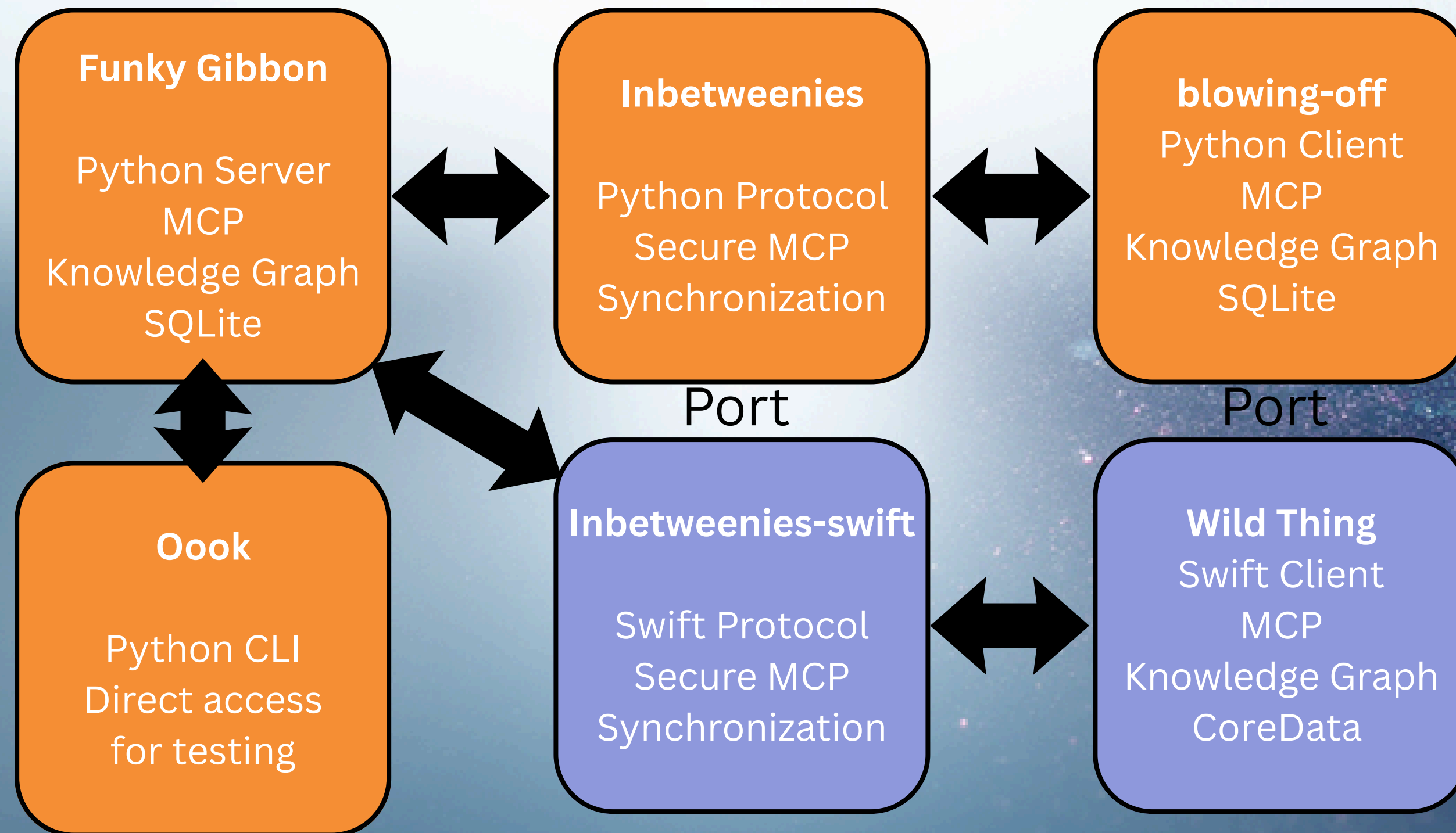
No backend as yet...



Distributed Portable MCP Service - The Goodies

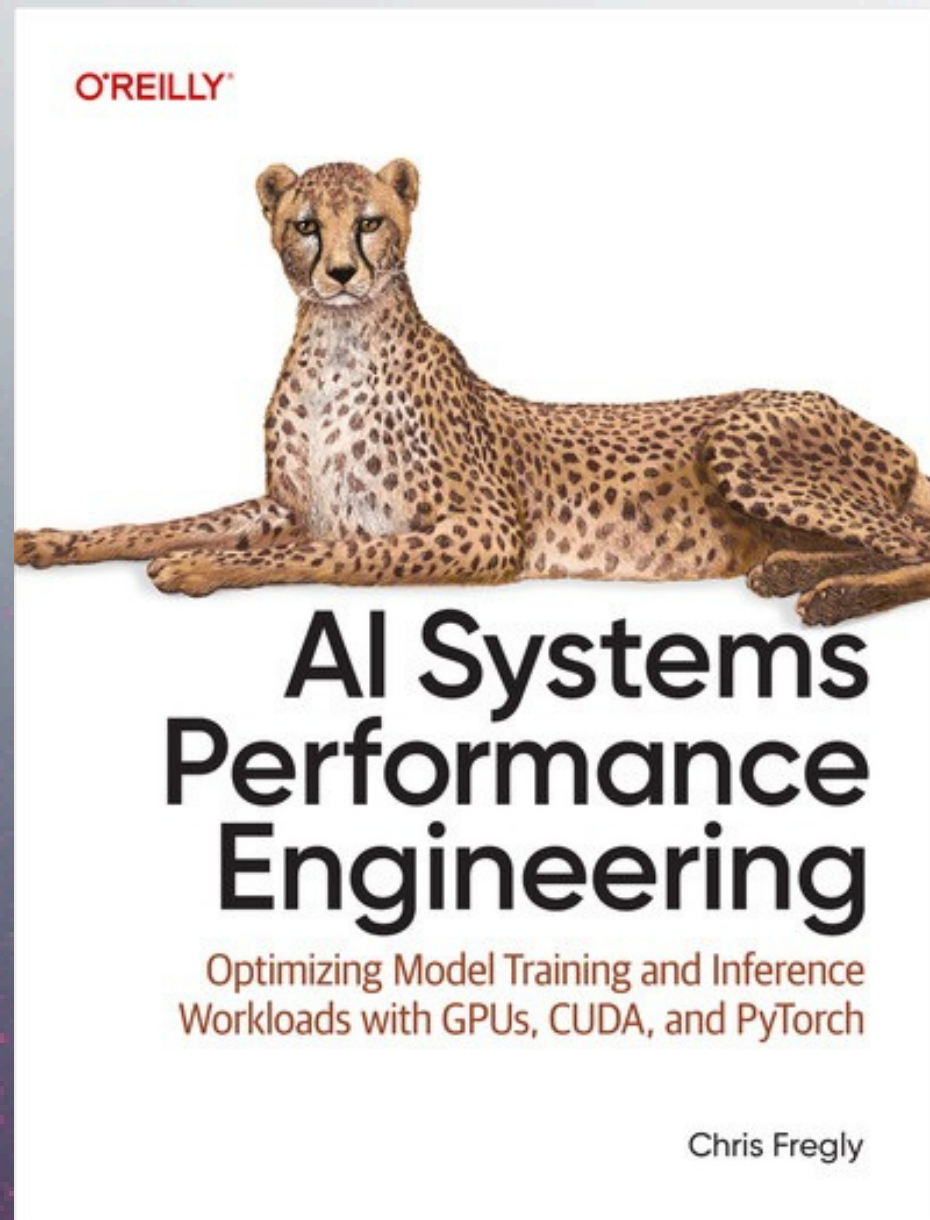
<https://github.com/adrianco/the-goodies> - structured high quality clean code. All 225 tests pass.

Security: JWT authentication, rate limiting, audit logging. Access Control: Admin and guest role QR code access



Work
in
Progress

Re-coding a big project from TypeScript to Python



I was a reviewer of this excellent book by Chris Fregly that was released Nov 2025.

A startup I'm working with wanted a demo version of their app idea, and Chris built it over a week or two using Cursor, mostly in TypeScript. I provided a MCP knowledge graph component in Python.

I later used claude-flow hive-mind to rewrite it all in Python starting with the BDD tests, using the tests to drive the rewrite to completion.

Demo Agentic Coding at Nubank - This is all I said



Adrian: If I wanted to create an MCP knowledge graph for Brazilian soccer players, teams and games, is the data available to populate the graph?

Claude:... <too much output to show>

Adrian: don't need real time, need free access, this is for a demo, not commercial use

Claude:... <too much output to show>

Adrian: I will want to connect the MCP server to Claude and have a set of sample questions for the demo that Claude looks up via MCP to come up with the answers

Claude:... <too much output to show>

Adrian: save this entire conversation as a document that I can implement elsewhere



Test Questions from the doc Claude wrote

Simple Lookups:

1. "Who scored the most goals for Flamengo in 2023?"
2. "What teams has Neymar played for in his career?"
3. "When was the last match between Palmeiras and São Paulo?"

Relationship Queries:

4. "Which players have played for both Corinthians and Palmeiras?" (rival teams)
5. "Show me all Brazilian players who moved to European clubs in 2024"
6. "What teams did Ronaldo Nazário play for during his career?"

Aggregate/Analysis:

1. "Which team has won the most Brasileirão titles?"
2. "Who are the top 5 goal scorers in Brazilian Serie A history?"
3. "What's Flamengo's win rate against Internacional in the last 10 years?"

Complex Multi-hop:

4. "Find players who scored in a Copa do Brasil final and later played in Europe"
5. "Which coaches have managed multiple championship-winning teams?"
6. "Show me teammates of Neymar at Santos who also made it to European leagues"



Pre-Work

Created a new public Github Repo

<https://github.com/adrianco/demo-brazil> (now GitHub.com/brazil-bench)

Started a Codespace

Installed Claude-Code and Claude-Flow

Copied the document to the repo

Installed Neo4j

Now it's YOLO-demo-Time!



I ran this command to see what happened

```
$ npx claude-flow@alpha swarm "Read brazilian-soccer-mcp-guide.md and implement phases 1,2 and 3 as described and test using BDD GWT structured PyTest. Use Neo4j as documented in docs/neo4j-setup.md, maintain a detailed context block comment at the start of every code file. Finally update README.md to describe what was done and push everything to github" --claude
```

It took about an hour elapsed time, then I tinkered with it for another hour



Swarm to-do list

prompts.txt

```
342
343 * Initializing swarm coordination for Brazilian Soccer MCP_ (esc to interrupt · ctrl+t to show todos)
344 |   Next: Read and analyze Phase 1 requirements from guide
345
346 |   □ Initialize swarm coordination for Brazilian Soccer MCP
347 |   □ Read and analyze Phase 1 requirements from guide
348 |   □ Read and analyze Phase 2 requirements from guide
349 |   □ Read and analyze Phase 3 requirements from guide
350 |   □ Set up project structure and directories
351 |   □ Create Neo4j connection module
352 |   □ Implement data loader for Kaggle datasets
353 |   □ Create graph schema and entities
354 |   □ Implement MCP server with core tools
355 |   □ Write BDD tests using pytest-bdd
356 |   □ Test Phase 1 implementation
357 |   □ Test Phase 2 implementation
358 |   □ Test Phase 3 implementation
359 |   □ Update README.md with documentation
360 |   □ Commit and push to GitHub
361
```

Brazilian Soccer MCP Benchmark

Run a swarm with the same objective to see how the swarm results evolve...

Setup as Github organization [GitHub.com/brazil-bench](https://github.com/brazil-bench) with bench template
Attempts are evaluated, compared, issues filed using Pourpoise SOP scripts
There are 9 attempts in the repo, including Claude-flow, Beads and Gastown

Top 10 Leaderboard								
Rank	Attempt	Pattern	Score	Spec	Tests	Integration	Skip%	Issues
1 🏆	2025-12-14-python-claude-beads-2	Beads v3	91.8	16/16	59	× External	20%	2 open
2 🥈	2025-10-30-python-hive	Hive v1	90.9	15/16	64	× External	0%	3 open
3 🥉	2025-12-15-python-claude-ruvector	RuVector	90.1	16/16	61	× External	0%	2 open
4	2025-12-13-python-claude-hive	Hive v2	86.0	16/16	43	✓ testcontainers	0%	0
5	2026-01-13-python-gastown	Gastown	85.4	16/16	40	× External	0%	1 open
6	2025-09-30-python-swarm	Swarm v1	83.5	16/16	63	✓ testcontainers	0%	0
7	2025-12-13-python-claude-swarm	Swarm v2	82.8	16/16	37	× Skip	3%	1 open
8	2025-12-14-python-claude-beads	Beads v2	78.4	16/16	25	✓ testcontainers	0%	0
9	2025-12-01-python-claude-beads	Beads v1	75.5	16/16	18	× Mock	0%	1 open

- **Best Overall:** Beads v3 (100% compliance, 59 effective tests)
- **New Entry:** Gastown (#5) - fastest implementation (~18 min), zero fix commits
- **testcontainers leaders:** Hive v2, Beads v2, and Swarm v1 have self-contained integration tests
- **Most Tests:** Hive v1 (64), Swarm v1 (63) - both with 0% skip ratio
- **Full Compliance:** 8 of 9 attempts achieve 16/16 (Hive v1 at 15/16)
- **Cleanest Code:** Gastown (0 fix commits, 40 tests pass first try)
- **6 of 9 fail integration:** 3 use testcontainers (Hive v2, Beads v2, Swarm v1)

What's Missing?

I'm doing too many repetitive mindless management tasks

“Make 100% of the tests pass”, “Tidy first”,

“Yes, keep going until you have finished”, “Archive old docs”,

“No don't skip that bit that didn't work”, “Push changes to github”,

“Compare the Python code functionality with the Typescript code again”

Need a Director agent. The Hive Queen is more of a line manager (GT Mayor?)

Really, need a complete systemic platform for managing agent development

Break down tasks, manage to guardrails and policies

Automating the need for Development Management away... (perhaps NoMan?)

Start with simple Standard Operating Procedure based purpose management

References

<https://www.anthropic.com/claude-code>

<https://github.com/ruvnet/claude-flow>

Adrian's swarm coding experiments since June 22nd, 2025 - making my house self-aware

<https://github.com/adriano/consciousness> - first try, did too much with claude-flow in one go

<https://github.com/adriano/c11s-house-ios> - built a Swift/iOS app from scratch

<https://github.com/adriano/the-goodies> - distributed MCP knowledge graph in Python

Brazilian soccer

<https://github.com/brazil-bench> - Framework benchmarking and code review work

<https://adriano.medium.com> - Blog posts

<https://linkedin.com/in/adriano-cockcroft> - Discussion about agentic coding

The background is a vibrant, abstract composition of various geometric shapes in shades of blue and orange. A large, bright white circle is centered on the page, serving as a backdrop for the text. Surrounding this circle are numerous 3D-style shapes, including circles, squares, rectangles, and question marks, some of which are scattered with fine orange dust or particles. The overall aesthetic is modern and creative.

YMMV: Questions?

**Ask: Join Brazil-bench
and contribute attempts**